

Coding & Solution

Date	23 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Solution Summary

Field	Details
Repository Link / URL	https://github.com/venkatasivasaitadiparthi-lgtm/Rising-Waters-Flood-Prediction.git
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn, XGBoost
Key Features Implemented	• Historical flood dataset loading and preprocessing • Exploratory Data Analysis (EDA) • Missing value handling and outlier treatment using IQR • Feature scaling using StandardScaler • Machine Learning model training (Decision Tree, Random Forest, KNN, XGBoost) • Model evaluation using Accuracy Score, Confusion Matrix, and Classification Report • Best model selection • Model serialization using Joblib (<code>floods.save</code>, <code>scaler.save</code>) • Flask web application • User-friendly HTML/CSS interface • Real-time flood prediction
Pending / Incomplete Features	• Real-time weather API integration • Cloud deployment (IBM Cloud/AWS) • SMS/Email flood alerts • Interactive dashboard and GIS mapping
Setup / Run Instructions	1. Install Python and required libraries using <code>pip install -r requirements.txt</code>. 2. Place <code>floods.save</code> and <code>scaler.save</code> in the project folder. 3. Open the project folder in Visual Studio Code. 4. Run <code>python app.py</code>. 5. Open <code>http://127.0.0.1:5000</code> in a web browser. 6. Enter weather and rainfall values and click Predict Flood to view the prediction.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Rising Waters project implements a complete machine learning pipeline for flood prediction, including data preprocessing, exploratory data analysis, model training, evaluation, and deployment through a Flask web application. The system allows users to enter weather and rainfall parameters through a web interface and instantly predicts whether there is a Flood Chance or No Flood Chance using the trained machine learning model. The project is designed as an educational decision-support system for disaster preparedness and demonstrates the practical application of machine learning in environmental risk assessment.